Calibration After Extending an LLM Vocabulary with Geometry-Aware Domain Tokens

Technical note for discussion and collaboration

Domain examples: points of interest (POIs), drugs, molecular entities, or other structured domain identifiers

Main conclusion. Extending an LLM vocabulary with new domain-specific token embeddings does *not* automatically require a generic post-hoc calibration stage. If the embeddings are learned or initialized through a geometry-aware process, the central question is whether their **relative domain geometry** is also **compatible with the pretrained LLM embedding and logit geometry**. The recommended procedure is therefore: geometry-aware initialization/alignment, supervised fine-tuning, then explicit diagnostics of norm, manifold alignment, geometry preservation, and output-logit behavior. A separate calibration step is introduced only if those diagnostics reveal a systematic mismatch.

1 Problem setup

Let the pretrained model have vocabulary size V and embedding dimension d, with embedding matrix

$$E_{\text{base}} \in \mathbb{R}^{V \times d}. \tag{1}$$

Suppose we add K new domain-specific tokens, such as POI identifiers or drug identifiers, with embeddings

$$E_{\text{new}} \in \mathbb{R}^{K \times d}. \tag{2}$$

The extended embedding table is

$$E' = \begin{bmatrix} E_{\text{base}} \\ E_{\text{new}} \end{bmatrix} \in \mathbb{R}^{(V+K) \times d}. \tag{3}$$

In a simple vocabulary expansion, the new rows may be initialized randomly or from related text tokens and then learned during fine-tuning. In the setting considered here, the new embeddings are instead produced by a **geometry-aware process**: their pairwise relationships encode meaningful domain structure, for example geographic proximity, graph relations, molecular similarity, semantic similarity, or a combination of such signals.

The important point is that two different notions of correctness are now involved:

- **Domain geometry:** e_i and e_j should have the intended relative relationship.
- **LLM compatibility:** e_i should interact appropriately with pretrained hidden states and the output softmax.

A geometry-aware method can solve the first problem without necessarily solving the second.

2 Is post-fine-tuning calibration required?

Not necessarily. There is no universal rule that a model must be calibrated simply because its vocabulary has been extended. Fine-tuning can learn useful new embeddings directly, provided the new tokens receive sufficient training signal and the optimization is well behaved.

However, for structured domain tokens, it is useful to replace the vague question "Do we need calibration?" with a more precise question:

> *After geometry-aware initialization and fine-tuning, are the new tokens correctly positioned, scaled, and used relative to the pretrained representation and output spaces?*

If the answer is yes, no additional calibration stage may be needed. If not, targeted alignment or calibration can be added.

3 What a geometry-aware process already gives us

Assume a domain encoder or geometric procedure yields vectors

$$Z = \{z_1, \dots, z_K\}, \tag{4}$$

with a meaningful similarity or distance structure. A mapping f places these vectors in the LLM embedding space:

$$e_i = f(z_i). \tag{5}$$

The training procedure may explicitly preserve relations such as

$$d_{\text{domain}}(z_i, z_j) \approx d_{\text{LLM}}(e_i, e_j), \tag{6}$$

or corresponding cosine similarities.

This is valuable because the new tokens do not begin as unrelated arbitrary vectors. For example, two pharmacologically or structurally similar drugs can start near one another, or two POIs with related geographic/semantic properties can share local structure.

But pairwise geometry alone does not determine the absolute placement of the new embedding cloud. The entire cloud can preserve its internal shape while having the wrong mean, norm, scale, orientation, or covariance relative to the pretrained LLM space. This motivates the following diagnostics.

4 Recommended diagnostics

4.1 Embedding norm compatibility

Compare the norm distribution of the new tokens with that of relevant pretrained tokens:

$$\|e_{\text{new}}\| \quad \text{versus} \quad \|e_{\text{base}}\|. \tag{7}$$

A useful first-order check is

$$\mathbb{E}[\|e_{\text{new}}\|] \approx \mathbb{E}[\|e_{\text{base}}\|]\,, \tag{8}$$

while also comparing variances and tails rather than only the mean.

A systematic norm mismatch can matter strongly when embeddings participate in the language-model output head. If a hidden state is h, an output logit often contains a dot product of the form

$$\ell_i = h^\top e_i, \tag{9}$$

so unusual embedding norms can create systematic logit biases.

4.2 Manifold or space alignment

The domain geometry should be anchored to concepts that the pretrained LLM already understands. For example, a drug identifier could be aligned with an embedding derived from the drug name, description, indications, or molecular semantics; a POI identifier could be aligned with textual or semantic descriptions such as category, neighborhood, geographic context, or attributes.

A simple alignment model is

$$e_i = Wz_i + b. \tag{10}$$

If preservation of the original geometry is especially important, W can be regularized or constrained, for example toward an orthogonal or Procrustes-style transformation. More flexible nonlinear mappings may be appropriate when the domain geometry and pretrained LLM geometry are not approximately related by a rigid transform.

The key distinction is:

$$\text{good internal geometry} \;\not\Rightarrow\; \text{good absolute placement in the LLM space.} \tag{11}$$

4.3 Output-logit calibration and competition

For tokens that the model must **generate**, this is one of the most important checks. Given representative hidden states h from the target tasks, evaluate the logits of new and old vocabulary items:

$$\ell_i = h^\top e_i \quad \text{(for tied embeddings)}, \tag{12}$$

or the corresponding output-head weights in an untied model.

Compare quantities such as

$$\mu_{\text{new}} = \mathbb{E}_{h,i}\left[h^\top e_i\right], \tag{13}$$

$$\sigma_{\text{new}} = \operatorname{Std}_{h,i}\left[h^\top e_i\right] \tag{14}$$

with analogous statistics for appropriate subsets of the original vocabulary.

The goal is not to force identical distributions. The goal is to detect pathologies such as:

- new tokens almost never receiving competitive logits;
- new tokens receiving abnormally large logits in unrelated contexts;

- systematic bias caused by norms or placement rather than task evidence;
- excessive probability mass being transferred from the original vocabulary.

When generation of domain IDs is a primary task, evaluate ranking metrics directly as well: top-k recall, mean reciprocal rank, accuracy, calibration curves over candidate entities, and probability mass assigned to valid versus invalid domain tokens.

4.4 Geometry preservation after fine-tuning

A geometry-aware initialization can be partially or completely destroyed by supervised fine-tuning. Therefore compare the domain geometry before and after SFT.

For cosine geometry, one can examine

$$G_{\text{before}}(i,j) = \cos(e_i, e_j), \tag{15}$$
$$G_{\text{after}}(i,j) = \cos(e'_i, e'_j). \tag{16}$$

Alternatively, compare full pairwise distance matrices or neighborhood graphs. Useful summary measures include rank correlation of pairwise similarities, neighborhood overlap, trustworthiness/continuity, or task-specific structural metrics.

If the geometry degrades substantially, it can be preserved during SFT with a regularizer:

$$\mathcal{L} = \mathcal{L}_{\text{SFT}} + \lambda \mathcal{L}_{\text{geo}}, \tag{17}$$

where, for example,

$$\mathcal{L}_{\text{geo}} = \sum_{i,j} \left(\cos(e_i, e_j) - S_{ij}^{\text{domain}} \right)^2. \tag{18}$$

This is often preferable to allowing SFT to destroy the desired structure and attempting to repair it afterward.

5 Input-side versus output-side domain tokens

The importance of calibration depends strongly on how the new tokens are used.

Use of new tokens	**Primary concern**	**What to emphasize**
Input-side only	Whether the token representation supplies useful domain information to the transformer	Manifold alignment, semantic anchoring, geometry preservation, downstream task performance
Generated as output IDs	Whether new tokens compete correctly in the softmax and are selected in the right contexts	All input-side checks plus logit distributions, ranking behavior, candidate normalization, and output calibration

If the model uses **tied input and output embeddings**, the extended embedding matrix also participates directly in the output vocabulary. If the language-model head is **untied**, the output projection must also be extended and trained:

$$W_{\mathrm{LM}} \in \mathbb{R}^{V \times d} \quad \longrightarrow \quad W'_{\mathrm{LM}} \in \mathbb{R}^{(V+K) \times d}. \tag{19}$$

Without this, the model may consume the new tokens as inputs but cannot properly predict them as outputs.

6 Probability calibration is a separate question

The discussion above concerns **representation and logit compatibility**. This is distinct from classical probability calibration, where a confidence of 0.8 is expected to correspond to approximately 80% empirical correctness.

If the application requires well-calibrated probabilities, post-hoc methods such as temperature scaling may still be useful:

$$p_i(T) = \frac{\exp(\ell_i/T)}{\sum_j \exp(\ell_j/T)}. \tag{20}$$

This should be decided based on held-out calibration error or task requirements, not merely because vocabulary extension occurred.

For entity generation, global temperature scaling may also be too coarse. Depending on the application, it can be useful to evaluate calibration separately for original tokens versus domain tokens, or within a constrained candidate set of valid entities.

7 Recommended pipeline

For geometry-aware POI, drug, or similar domain tokens, a practical pipeline is:

1. **Construct domain geometry.** Learn or define vectors/relations using the domain-specific geometric process.
2. **Align to the LLM space.** Use semantic anchors, textual descriptions, shared entities, or a learned mapping so the new cloud is not only internally coherent but also sensibly positioned relative to pretrained representations.
3. **Check initial scale and norms.** Verify that norms and basic statistics are not pathological relative to relevant pretrained tokens.
4. **Extend the input embedding table and, when necessary, the output head.**
5. **Fine-tune on representative tasks.** Ensure the new tokens occur sufficiently often and in contexts that teach the intended behavior.
6. **Preserve geometry during SFT if it is part of the inductive bias.** Use freezing, partial freezing, smaller learning rates, or an explicit geometry regularizer as appropriate.
7. **Evaluate after SFT.** Measure domain geometry, manifold alignment, token norms, downstream performance, and – for generated tokens – logit/ranking behavior.

8. **Calibrate only if needed.** Add a targeted correction (e.g., rescaling, alignment, bias correction, temperature scaling, or candidate-level calibration) when the held-out diagnostics identify a systematic mismatch.

A compact summary is

$$\begin{aligned} &\text{domain geometry} \rightarrow \text{LLM-space alignment} \rightarrow \text{norm check} \\ &\qquad \rightarrow \text{SFT} \rightarrow \text{geometry} + \text{logit diagnostics.} \end{aligned} \tag{21}$$

8 Interpretation for POIs and drugs

The same framework applies to both examples, but the geometry being preserved may differ.

POIs. The geometry may combine geographic distance, category structure, mobility/co-visit patterns, neighborhood context, textual descriptions, or graph connectivity. Pure geographic geometry should not automatically be assumed to match semantic geometry in the LLM, so semantic anchors are especially useful.

Drugs. The geometry may encode molecular structure, targets, indications, adverse-effect profiles, interaction graphs, or biomedical-text similarity. Again, preserving this geometry is beneficial, but the embedding cloud still needs to be aligned with the pretrained model's biomedical or textual concepts if the model is expected to reason jointly over domain IDs and natural language.

In both cases, if the new tokens represent *entities* rather than conventional linguistic subwords, the objective is not necessarily to make them statistically indistinguishable from ordinary vocabulary embeddings. The objective is to make them **usable by the transformer** while preserving the domain structure that motivated their introduction.

9 Bottom line

For geometry-aware domain vocabulary expansion, calibration is best treated as an **empirical compatibility question**, not as a mandatory final stage.

- If domain geometry is preserved, LLM-space alignment is good, and new-token logits behave appropriately, **no extra calibration step is required**.
- If the embeddings have good internal geometry but are mis-scaled or poorly placed relative to the pretrained space, use **alignment or norm/logit correction**.
- If SFT destroys the intended structure, prefer **geometry-preserving training or regularization** rather than relying only on post-hoc repair.
- If calibrated probabilities are an application requirement, treat **probability calibration** as a separate held-out evaluation problem.

The most consequential distinction is whether the new domain tokens are only consumed as inputs or are also generated as outputs. For input-side tokens, geometric and semantic alignment usually dominate. For generated entity tokens, output-logit competition and candidate-level calibration deserve substantially more attention.